Per-Layer Lookahead Masks Leak Future Context: Re-measuring the Future-Context Requirement of Streaming Accent Conversion

Dipesh Tripathi

***Abstract*—Streaming foreign accent conversion systems are designed around a lookahead budget, and studies of the latency–quality trade-off report accuracy as a function of that budget. Such results are valid only if the model's output depends on no more future input than the budget states. We show that two common constructions violate this. First, masking self-attention does not make wav2vec 2.0 or WavLM base causal: a positional convolution admits 1.28 s of future input and a feature-encoder normalisation admits the whole utterance. Second, a lookahead mask applied in every layer compounds with depth: in a 13-layer model built on a frozen WavLM encoder, a nominal 40 ms budget yields at least 220 ms of effective future context, and a nominal 640 ms at least 2.8 s. We give a perturbation test that measures effective future context and two masking schemes for which it equals the nominal budget exactly. Re-measuring a phonetic-normalisation probe for accent conversion under these schemes changes the conclusion substantially. Under per-layer masking, phone error rate falls by 63% from 0 to 640 ms of nominal lookahead. With exact budgets the reduction is 12.5% when the lookahead enters the first encoder layer and 3.7% when it enters only after the encoder, where nearly all of the gain is obtained within 80 ms. Two secondary findings of the per-layer analysis, a stronger lookahead response for canonical than for realized phone targets and a large difference between encoders, do not survive exact budgets. We conclude that lookahead results should be reported with a measured effective context.**

***Index Terms*—accent conversion, streaming speech processing, lookahead, latency, self-supervised models, reproducibility**

I. Introduction

Streaming foreign accent conversion (FAC) has been demonstrated at low latency. PHONOS [2] operates with at most 40 ms of lookahead and reports an 81% reduction in non-native accent confidence, and related systems for streaming conversion and anonymisation operate under similar budgets [3], [4]. The lookahead budget is therefore a primary design parameter of such systems, and a natural question is how much accuracy each additional millisecond of future context buys.

Answering that question requires that the budget a model is labelled with is the future context it actually uses. Two properties of current practice make this assumption fragile. Self-supervised speech encoders contain components outside self-attention that read future input, so masking attention alone does not make them causal. And a lookahead mask applied identically in every layer of a deep network extends the receptive field by the budget once per layer. The latter is known in streaming speech recognition, where the look-ahead latency of per-layer right context is recognised to aggregate over the layers of the stack [5]. It is easily overlooked, however, when a frozen pretrained encoder is masked to emulate a streaming condition, because the nominal budget then describes the mask rather than the model.

This paper began as a characterisation of the lookahead–accuracy trade-off for streaming FAC using a phone-translation probe, and an earlier version reported a steep, log-linear benefit of lookahead under per-layer masking. A measurement of the probe's effective future context showed that the masks compounded and that the nominal budgets understated the context used by a factor of three to five at short budgets and more at long ones. The present paper reports that finding, the tools that detect and remove it, and the corrected measurement.

The contributions are as follows.

1) We identify two components of wav2vec 2.0 and WavLM base that leak future input under attention masking, give patches for both, and verify them by a truncation test (§III-D).
2) We show that per-layer lookahead masks compound in a frozen-encoder probe and quantify the inflation, using a perturbation test that measures the effective future context of a complete model (§III-B, §V-A).
3) We define two masking schemes whose effective context equals the nominal budget exactly, one placing the lookahead after the encoder and one in the first encoder layer, and verify both with the same test.
4) We re-measure the lookahead response of the probe under all three schemes with matched training, and show that the per-layer result overstates the benefit of lookahead by a factor of five to seventeen and that two secondary findings derived from it do not hold under exact budgets (§V-B–§V-E).

II. Related work

Nguyen et al. [6] present a streaming non-autoregressive accent conversion system built on an Emformer encoder, and PHONOS [2] targets substantially lower latency with bounded future context. PHONOS, TVTSyn [3] and DarkStream [4] originate from one group and combine conversion with speaker

D. Tripathi is with the University of South Dakota, Vermillion, SD, USA (e-mail: dipesh.tripathi@coyotes.usd.edu).

Code, causality and effective-context tests, sweep configurations, analysis scripts and per-condition results: https://github.com/Dipeshtripathi13/streaming-fac-lookahead. Data is L2-ARCTIC [1] via `KoelLabs/L2Arctic`. Given that corpus's CC-BY-NC-4.0 licence, configurations rather than model weights are released.

anonymisation. TVTSyn reports both CPU and GPU latency at a 60 ms chunk, and DarkStream ablates its look-ahead over $\{0, 140, 280\}$ ms, selecting 140 ms because the further 140 ms improves anonymisation accuracy by less than one percentage point. StreamVC [7] reports mobile operation without device details. A survey [8] frames the problem space; streaming anonymisation [9], [10], [11] shares the latency constraint, and offline FAC [12], [13] and low-latency voice conversion [14] bracket the design space.

In streaming speech recognition, Zhang et al. [5] note that applying the same right-context mask in every Transformer layer makes the look-ahead latency aggregate over the layers. Our contribution is not that observation but its measurement and consequences in a setting where it is easy to miss: a frozen self-supervised encoder masked to emulate streaming, used as a probe of how much future context a conversion task requires. None of the FAC systems above is evaluated here, and no claim is made about their effective context.

III. Method

A. Task and latency model

Let $\mathbf{x}$ be a 16 kHz waveform from a non-native speaker, encoded into frames $\mathbf{z}_{1:T}$ at hop $h = 20$ ms. L2-ARCTIC [1] annotates each utterance with a canonical phone sequence $\mathbf{y}^{\mathrm{can}}$ derived from the lexicon and a realized sequence $\mathbf{y}^{\mathrm{real}}$ transcribed from what the speaker produced. We operationalize the phonetic-normalisation component of accent conversion as the prediction of $\mathbf{y}^{\mathrm{can}}$ from accented audio,

$$f_\theta : \mathbf{x} \longmapsto \mathbf{y}^{\mathrm{can}}. \tag{1}$$

This is a controlled proxy for one stage of FAC; the future-context requirements of acoustic generation and prosody may differ.

A streaming system consumes audio in chunks of $c = Ch$ and may delay its output by a lookahead of L, realized as $N_L = \lceil L/h \rceil$ frames. Under a block-emission convention its end-to-end delay is

$$t_{\mathrm{e2e}} = \underbrace{c + L}_{t_{\mathrm{alg}}} + t_{\mathrm{cmp}} + t_{\mathrm{buf}}, \tag{2}$$

where t_{cmp} is per-chunk computation and t_{buf} is input/output and jitter buffering. Eq. 2 holds only if the output for a chunk depends on no input beyond L after the end of the chunk. Verifying that condition is the subject of the following subsections.

B. Lookahead masking schemes

With zero-based frame indices, chunk length C, lookahead N_L and lookback N_B frames, the chunked lookahead mask permits query frame i to attend to key frame j if and only if

$$i - N_B \;\le\; j \;\le\; (\lfloor i/C \rfloor + 1)\, C - 1 + N_L. \tag{3}$$

The upper limit is the last frame of the chunk containing i, extended by N_L frames. The mask is added to the pre-softmax attention logits; in WavLM it is added to the gated relative position bias, which leaves the model's own padding mask in effect. All experiments use $C = 2$ ($c = 40$ ms) and $N_B = 100$ (2 s). We compare three ways of applying it to a model consisting of a frozen encoder tapped at layer ℓ followed by a four-block probe.

Per-layer. Eq. 3 with lookahead N_L is applied in every encoder layer and every probe block. The bounds compose: a frame at layer k attends to frames of layer $k-1$ that have themselves attended N_L frames ahead, so after K masked layers the output at frame i can depend on input as far as approximately $i + K(N_L + C - 1)$. For $N_L = 0$ the bound does not grow, since every attended frame lies in the same or an earlier chunk. For $N_L > 0$ it grows linearly with depth.

End-to-end. The encoder is masked frame-causally (Eq. 3 with $C = 1$ and $N_L = 0$), the first probe block uses the full N_L, and the remaining blocks use $N_L = 0$. An encoder frame then depends on no input beyond itself, the first block extends each frame's context to the end of its chunk plus N_L frames, and later blocks, which attend only within the current and earlier chunks, cannot extend it further. The effective future context is exactly $(C - 1 + N_L)h$.

Encoder-first. The first encoder layer uses the full N_L and every later encoder layer and probe block uses $N_L = 0$. By the same argument the effective future context is again exactly $(C - 1 + N_L)h$, but the pretrained encoder now builds its representation with the future in view.

The two exact schemes differ in a second respect that matters at $L = 0$: in the end-to-end scheme encoder frames are frame-causal, whereas in the encoder-first and per-layer schemes they may also attend to the remainder of their own chunk.

C. Measuring effective future context

Let F denote the complete model, encoder and probe. We replace every sample after a time τ by independent noise and locate the earliest output frame whose relative change, $\|F(\mathbf{x})_t - F(\mathbf{x}')_t\| / \|F(\mathbf{x})_t\|$, exceeds 10^{-4}; floating-point noise in this computation is of order 10^{-5}. The interval between that frame and the first input frame affected by the perturbation is the effective future context. Because a frame's context depends on its position within the chunk, the test is repeated for each alignment of τ within a chunk and the largest value is retained. The probe is randomly initialised, since the property under test is structural. Values obtained this way are lower bounds, because the influence of sufficiently distant frames can fall below the threshold. The test is run for every budget before each training sweep, and the sweep is aborted if an exact scheme fails it.

D. Making the encoder causal

Masking self-attention in wav2vec 2.0 or WavLM base [15], [16] does not produce a causal encoder, because two components outside the attention layers also read future input.

Positional convolution. The convolutional positional embedding is a grouped convolution with kernel size $K_p = 128$ and symmetric padding, so the embedding added to frame t is computed from frames $t-64$ to $t+63$, 1.28 s of future context.

TABLE I
TRUNCATION TEST ON `WAVLM-BASE-PLUS` AT $N_L = 0$: RELATIVE L_2 CHANGE δ IN EARLIER FRAMES AFTER LATER AUDIO IS DELETED.

configuration	δ	causal?
attention mask only	1.14×10^{-2}	no
+ causal positional conv	6.00×10^{-3}	no
+ cumulative normalisation	2.60×10^{-2}	no
both patches	$\mathbf{4.58 \times 10^{-6}}$	**yes**

We replace the symmetric padding with $K_p - 1$ frames of left padding,

$$\mathbf{p}_t = \sum_{k=0}^{K_p-1} \mathbf{W}_k \, \mathbf{z}_{t-(K_p-1)+k}, \tag{4}$$

retaining the pretrained kernel $\mathbf{W}$.

Feature-encoder normalisation. In the base checkpoints the first layer of the convolutional feature encoder is followed by a group normalisation with one group per channel, whose statistics are computed over the entire utterance. We replace it with a cumulative normalisation,

$$\begin{aligned} \mu_t &= \frac{1}{t}\sum_{s=1}^{t} x_s, \qquad \sigma_t^2 = \frac{1}{t}\sum_{s=1}^{t} x_s^2 - \mu_t^2, \\ y_t &= \gamma \, \frac{x_t - \mu_t}{\sqrt{\sigma_t^2 + \epsilon}} + \beta, \end{aligned} \tag{5}$$

applied per channel with the pretrained affine parameters γ and β. The large checkpoints use layer normalisation at this position and do not require the second patch. Both patches alter the function computed by the pretrained encoder, which is then used frozen; every condition uses the same patched encoder, so comparisons across lookahead are unaffected.

Verification by truncation. A causal encoder must satisfy $F(\mathbf{x})_t = F(\mathbf{x}_{1:\tau})_t$ for every frame t ending before sample τ. With $N_L = 0$, a 4 s input truncated at 2 s and the last $m = 8$ frames excluded to cover the front end's receptive field, we compute the relative L_2 change δ in the remaining frames and require $\delta < 10^{-4}$. Table I reports the four combinations of the two patches. Neither patch suffices alone, and applying the cumulative normalisation without the causal positional convolution increases δ. The value for the fully patched encoder lies at single-precision accumulation level and varies with the library release (4.58×10^{-6} here, 6.14×10^{-6} with an earlier release).

The truncation test establishes causality at $N_L = 0$ only. It cannot detect the compounding of §III-B, which appears only for $N_L > 0$; that is the role of the test in §III-C.

E. Probe and matched control

The probe operates on the hidden states of encoder layer ℓ ($\ell = 9$ for WavLM base+, $\ell = 2$ for wav2vec 2.0 base; §V-E). It consists of four pre-norm blocks of width 768, each applying eight-head self-attention, a depthwise convolution of kernel size 31 with left-only padding followed by a pointwise convolution, and a feed-forward layer of width 2048 with GELU activation, all with residual connections and dropout 0.1. A final layer normalisation and a linear layer produce posteriors over 37 symbols (36 phones after stress marks are removed, and the CTC blank). The probe has 24.6 M parameters, is trained with the connectionist temporal classification objective, and is decoded greedily. Padded positions are zeroed after every sub-layer.

Replacing $\mathbf{y}^{\mathrm{can}}$ in Eq. 1 with $\mathbf{y}^{\mathrm{real}}$ yields a realized-phone transcription model that shares architecture, capacity, vocabulary, data order, seed and step count with the canonical-phone model and differs only in its target tensor.

F. Reported quantities

PER is the Levenshtein distance between hypothesis and reference divided by the reference length, averaged over test utterances. The *exchange rate* is the slope of the least-squares fit $\mathrm{PER}(L) = \alpha + \beta \log_2 L$ over $L \geq 20$ ms, in PER per doubling. The *repeatability floor* 2σ is twice the standard deviation of test PER over complete reruns of a condition at identical seed and configuration; differences below it are treated as unresolved. The *preference margin* of a model is its PER against the other target minus its PER against its own.

IV. EXPERIMENTAL SETUP

Data. We use the `KoelLabs/L2Arctic` release of L2-ARCTIC [1], 3599 phoneme-annotated utterances from 24 speakers of six native languages, excluding utterances shorter than 0.8 s or longer than 10 s. The data are divided into speaker-disjoint sets stratified by native language: one speaker per language for validation, one for testing, and the remaining speakers for training. Test PER is computed on a fixed subset of 400 test utterances.

Training. The encoder is frozen. The probe is trained for 1200 steps with batch size 8, AdamW (weight decay 0.01), a one-cycle schedule with peak learning rate 3×10^{-4} and gradient clipping at 1.0. The order of training utterances is identical across conditions. All training ran on an NVIDIA T4.

Sweeps. Under the per-layer scheme: a dense sweep of 21 budgets from 0 to 640 ms with two seeds on the canonical-phone arm, 16 of those budgets on the realized-phone arm, and a seven-budget wav2vec 2.0 sweep. Under the end-to-end scheme we repeat each of these and add a seven-budget, two-target, three-seed sweep, a repeatability experiment and a convergence check to 6000 steps. Under the encoder-first scheme we run seven budgets $\{0, 20, 40, 80, 160, 320, 640\}$ ms with two seeds for WavLM and one for wav2vec 2.0. Every sweep verifies that no two conditions share the same effective budget.

Repeatability floor. Repeating four budgets ($L \in \{40, 160, 240, 320\}$ ms) four times each at identical seed and configuration gives $\sigma = 0.00146$ ($2\sigma = 0.0029$) under the per-layer scheme and $\sigma = 0.00251$ ($2\sigma = 0.0050$) under the end-to-end scheme, each with 12 degrees of freedom. We use the floor of the scheme being analysed.

Compute. Per-chunk compute was measured on an Apple M4 (four performance and six efficiency cores, 16 GB) and an ARM64 Linux host (four cores, 3.8 GB), with 28 to 50 timed repetitions after three warm-up iterations.

TABLE II
EFFECTIVE FUTURE CONTEXT (MS) OF THE COMPLETE MODEL, WAVLM BASE+ TAPPED AT LAYER 9 WITH A FOUR-BLOCK PROBE, $c = 40$ MS. THE NOMINAL VALUE IS THE REMAINDER OF THE CHUNK PLUS L. PER-LAYER VALUES ARE LOWER BOUNDS AT A 10^{-4} THRESHOLD.

L (ms)	nominal	per-layer	end-to-end	encoder-first
0	20	20	20	20
20	40	≥200	40	40
40	60	≥220	60	60
60	80	≥360	80	80
160	180	≥780	180	180
640	660	≥2820	660	660

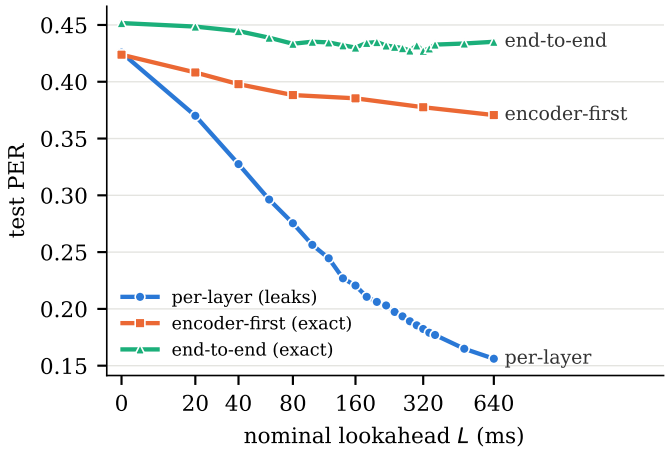


Fig. 1. Test PER of the canonical-phone probe (WavLM base+, mean of two seeds) against nominal lookahead under the three masking schemes. Only the two exact schemes have an effective future context equal to the nominal value (Table II).

V. RESULTS

A. Effective context of per-layer masks

Table II reports the effective future context of the complete WavLM model, nine encoder layers and four probe blocks, under the three schemes. Both exact schemes match the nominal value at every budget tested. Under the per-layer scheme the effective context is at least 200 ms at a nominal 20 ms, 220 ms at 40 ms, 780 ms at 160 ms and 2.82 s at 640 ms. The theoretical bound of §III-B, $13(N_L+1)$ frames, is 520 ms at a nominal 20 ms.

B. Lookahead response under the three schemes

Fig. 1 and Table III show test PER of the canonical-phone probe against nominal lookahead. Under the per-layer scheme PER falls from 0.4256 to 0.1561, a relative reduction of 63.3%, at an exchange rate of −0.0453 PER per doubling ($R^2 = 0.985$). Under the encoder-first scheme it falls from 0.4238 to 0.3707, a reduction of 12.5% at −0.0072 per doubling. Under the end-to-end scheme it falls from 0.4517 to a minimum of 0.4273 at 320 ms (5.4%) and ends at 0.4352 (3.7%), at −0.0033 per doubling ($R^2 = 0.60$). Relative to exact budgets, the per-layer analysis therefore overstates the endpoint benefit of lookahead by a factor of five against the encoder-first scheme and seventeen against the end-to-end scheme.

TABLE III
TEST PER (CANONICAL-PHONE ARM, WAVLM BASE+, MEAN OF TWO SEEDS) UNDER THE THREE SCHEMES, WITH THE EXCHANGE RATE (PER PER DOUBLING, $L \geq 20$ MS) AND THE RELATIVE REDUCTION FROM 0 TO 640 MS.

L (ms)	per-layer	encoder-first	end-to-end
0	0.4256	0.4238	0.4517
20	0.3700	0.4082	0.4486
40	0.3275	0.3979	0.4446
80	0.2754	0.3883	0.4335
160	0.2206	0.3854	0.4301
320	0.1823	0.3776	0.4273
640	0.1561	0.3707	0.4352
exchange rate	−0.0453	−0.0072	−0.0033
reduction 0–640	63.3%	12.5%	3.7%

End-to-end scheme in detail. The dense sweep samples every 20 ms to 360 ms. No individual 20 ms step exceeds the floor of 0.0050 (the largest, 40→60 ms, is 1.2× the floor), but the cumulative reduction from 0 to 80 ms is 0.0182 PER, 3.6× the floor and 4.0% relative. Beyond 80 ms the curve stays within about one floor of its 80 ms value and rises slightly between 360 and 640 ms. A three-seed sweep on seven budgets confirms the shape: the relative reduction from 0 to 640 ms is 2.8%, 4.4% and 4.8% across seeds.

Convergence. All conditions are trained for 1200 steps. To test whether the shallow response reflects under-training, five budgets were trained to 6000 steps under the end-to-end scheme. Test PER was 0.4183, 0.4078, 0.3941, 0.3892 and 0.4053 at 0, 40, 160, 240 and 640 ms. Longer training increases the benefit to 7.0% at 240 ms, but the response remains shallow and 640 ms is again worse than 240 ms.

Encoder-first scheme in detail. The two seeds give endpoint reductions of 14.2% and 10.8% and agree in shape, but differ by up to 0.024 PER at individual budgets (160 ms), about five times the end-to-end floor. The monotone decrease and the endpoint are well supported; differences between adjacent budgets above 80 ms are not resolved.

C. Where the context enters

Two comparisons indicate that the location of future context matters as well as its amount. First, at $L = 0$ the only difference between the encoder-first and end-to-end schemes is that encoder frames may attend to the remainder of their 40 ms chunk, one additional 20 ms frame for half of them. This alone reduces PER from 0.4517 to 0.4238, a 6.2% reduction, which exceeds the entire benefit of 640 ms of lookahead under the end-to-end scheme. Second, the encoder-first scheme continues to improve at 640 ms, whereas the end-to-end scheme is flat beyond 80 ms.

Amount nevertheless dominates the per-layer result. The per-layer model at a nominal 20 ms, with at most 520 ms of effective context, reaches 0.3700; the encoder-first model with a full 660 ms reaches 0.3707. The steep per-layer curve thus reflects effective contexts of several hundred milliseconds to several seconds distributed through all thirteen layers, which neither exact scheme provides at these budgets.

D. Canonical against realized targets

Under the per-layer scheme, the canonical-phone arm gained $1.55\times$ as much per doubling of lookahead as the realized-phone arm (-0.0452 against -0.0292), the PER gap between the arms widened from 0.017 at $L = 0$ to 0.094 at 640 ms, and the canonical arm's preference margin grew monotonically in three of three seeds. The earlier version took this as evidence that conversion requires more future context than accent-faithful transcription.

Under the end-to-end scheme the effect is not resolved. On the dense sweep the relative reduction from 0 to 640 ms is 3.7% for the canonical arm and 2.4% for the realized arm, with best reductions of 5.4% and 3.9%. On the three-seed sweep the per-seed reductions are 2.8, 4.4 and 4.8% for the canonical arm against 1.9, 3.5 and 7.8% for the realized arm, so the arms overlap. The canonical arm's preference margin changes from 0.029 at $L = 0$ to 0.033 at 640 ms and the realized arm's from 0.040 to 0.036; both changes are within the floor. We therefore withdraw the claim. Whether it holds under the encoder-first scheme, which was not run on the realized arm, remains open.

E. A second encoder

The wav2vec 2.0 base layer is selected by phone decodability at $L = 0$ under the end-to-end scheme: PER is 0.438, 0.465, 0.564, 0.699 and 0.775 at layers 2, 4, 6, 8 and 9, so layer 2 is used. Under the per-layer scheme this encoder appeared to respond very differently from WavLM, with a 3.9% endpoint reduction against 63.3%. Under that scheme, however, the two taps sit behind nine and two masked encoder layers respectively, so at equal nominal budgets they received very different effective context.

Under exact budgets the two encoders behave alike. In the end-to-end scheme, wav2vec 2.0 reaches its best PER at 80 ms, a 4.0% reduction from 0.4357 to 0.4183, and then degrades to 0.4544 at 640 ms; a second seed and a 6000-step run show the same rise beyond 160 ms. In the encoder-first scheme it is flat, with a best reduction of 1.6% (single seed). The large difference between encoders reported for the per-layer scheme was therefore mostly a difference in effective context.

F. Compute

Across $L \in [0, 640]$ ms at a 40 ms chunk, median per-chunk compute of the WavLM model varied by -0.3% to $+1.9\%$ on the M4 and by 0 to $+7.6\%$ on the ARM64 host, while the nominal algorithmic term varied seventeenfold. The throughput condition $\mathrm{RTF} = t_{\mathrm{cmp}}/c < 1$ is not met at this chunk size on the M4 ($t_{\mathrm{cmp}} \approx 43$–$44$ ms), independently of L. These measurements were taken under the per-layer scheme; the exact schemes apply a nonzero lookahead in fewer layers and are not expected to cost more.

VI. Measurement audits

Compounding lookahead. The central error of the earlier analysis is described above. It passed a truncation test, because that test ran at $N_L = 0$, where nothing compounds, and it produced a smooth and reproducible curve ($R^2 = 0.985$, replicated across seeds and a denser grid). Reproducibility of a curve says nothing about whether its abscissa is correct. The guard we now apply is to measure effective context at every budget before training.

A padding error changed the shape of the curve. An earlier version of the probe passed a zero-padded tensor to a depthwise convolution and feed-forward layer that ignore the padding mask. The resulting error grew with lookahead, from -0.014 PER at $L = 0$ to -0.086 at 320 ms, so it was not a constant offset. All results here postdate the correction.

A forced breakpoint estimator returns a breakpoint. A two-segment fit selected by ΔBIC [17] and a maximum-distance-to-chord estimator both reported confident knees on the per-layer curve, at 40 and 160 ms, although the curve is log-linear. On synthetic data that is log-linear for $L \geq 20$ ms with $L = 0$ off the extrapolation, including $L = 0$ yields $\Delta\mathrm{BIC} = -39$ and a knee at 40 ms, and excluding it yields $+5$. Identifiability must be tested by perturbing the axis.

Silent symbol mismatch. A phoneme-class map keyed on ARPAbet, applied to IPA output, sent 57% of reference tokens to an "other" class and still produced per-class results. A pre-specified manner-class analysis of the per-layer data, conducted after this was corrected, did not support the hypothesis that vowels and approximants benefit more from lookahead than obstruents. We do not repeat it under exact budgets, where the total benefit is only a few times the floor.

VII. Limitations

Phone-level probe. All quality results are phone error rates of a phone-translation probe over a frozen encoder, not perceived accentedness of converted speech, and the requirements of synthesis and prosody may differ.

Frozen encoder and scheme coverage. The patched encoder is used without adaptation. A fine-tuned encoder, or one trained for streaming, may exploit future context differently. The two exact schemes are two of many non-compounding designs; a budget distributed across layers, or a dedicated right-context block, was not tested.

Statistical resolution. Most sweeps use two seeds and the encoder-first wav2vec 2.0 sweep one. The repeatability floor is global, while the per-condition spread varies, and the encoder-first seeds disagree by up to five times the floor at individual budgets.

Effective-context bounds. Per-layer effective contexts are lower bounds at a fixed threshold; §V-C uses the theoretical upper bound where the measured value is not sufficient.

Second encoder. The layer selected for wav2vec 2.0 is shallow, leaving a single encoder layer above the encoder-first lookahead.

VIII. Conclusion

A lookahead budget describes a streaming model only if the model's output depends on no more future input than the budget states. In a frozen-encoder probe for streaming accent conversion, two conditions for this fail by default: masking attention leaves the encoder non-causal, and a per-layer lookahead mask compounds with depth, so that a nominal 40 ms

corresponded to at least 220 ms and a nominal 640 ms to at least 2.8 s. Both are detected by simple input-perturbation tests and both have direct remedies. With exact budgets, the probe's error rate falls by 12.5% over 640 ms when the lookahead enters the first encoder layer and by 3.7% when it enters after the encoder, with most of the latter obtained within 80 ms, against 63% under per-layer masking. Two secondary conclusions drawn from the per-layer analysis, a target asymmetry and an encoder dependence, do not survive. Studies of the latency–quality trade-off in streaming speech processing should report a measured effective context alongside the nominal budget.

Acknowledgements

This work received no external funding and used no institutional compute: GPU results are from Google Colab (Tesla T4) and CPU results from consumer hardware. Thanks to the maintainers of L2-ARCTIC and WavLM, whose public releases made a study of this scope possible.

Compliance with ethical standards

This work used only the publicly released L2-ARCTIC corpus and involved no new participant recruitment or data collection. No listening study was run and no claim about perceived accentedness is made. The author declares no competing interests.

References

[1] G. Zhao, S. Sonsaat, A. Silpachai, I. Lucic, E. Chukharev-Hudilainen, J. Levis, and R. Gutiérrez-Osuna, "L2-ARCTIC: A non-native English speech corpus," in *Interspeech*, 2018, pp. 2783–2787.

[2] W. Quamer, M.-R. Tseng, G. Nasrallah, and R. Gutiérrez-Osuna, "PHONOS: PHOnetic neutralization for online streaming applications," *arXiv preprint arXiv:2603.27001*, Mar. 2026.

[3] W. Quamer, M.-R. Tseng, G. Nasrallah, and R. Gutiérrez-Osuna, "TVTSyn: Content-synchronous time-varying timbre for streaming voice conversion and anonymization," *arXiv preprint arXiv:2602.09389*, Feb. 2026.

[4] W. Quamer and R. Gutiérrez-Osuna, "DarkStream: Real-time speech anonymization with low latency," in *Proc. IEEE ASRU*, 2025, pp. 1–7.

[5] Q. Zhang, H. Lu, H. Sak, A. Tripathi, E. McDermott, S. Koo, and S. Kumar, "Transformer transducer: A streamable speech recognition model with transformer encoders and RNN-T loss," in *Proc. IEEE ICASSP*, 2020.

[6] T.-N. Nguyen, N.-Q. Pham, Ş. Akti, and A. Waibel, "Streaming non-autoregressive model for accent conversion and pronunciation improvement," in *Interspeech*, 2025, pp. 4163–4167.

[7] Y. Yang, Y. Kartynnik, Y. Li, J. Tang, X. Li, G. Sung, and M. Grundmann, "StreamVC: Real-time low-latency voice conversion," in *Proc. IEEE ICASSP*, 2024, pp. 11 016–11 020.

[8] Y. Halychanskyi, J. S. Guo, and V. Kindratenko, "Accent conversion: A problem-driven survey of sociolinguistic and technical constraints," *arXiv preprint arXiv:2604.27281*, Apr. 2026.

[9] N. Kuzmin, S. Liu, K. A. Lee, and E. S. Chng, "Stream-voice-anon: Enhancing utility of real-time speaker anonymization via neural audio codec and language models," in *Proc. IEEE ICASSP*, 2026.

[10] N. Kuzmin, K. A. Lee, and E. S. Chng, "StreamVoiceAnon+: Emotion-preserving streaming speaker anonymization via frame-level acoustic distillation," *arXiv preprint arXiv:2603.06079*, Mar. 2026.

[11] W. Quamer and R. Gutiérrez-Osuna, "End-to-end streaming model for low-latency speech anonymization," in *Proc. IEEE SLT*, 2024, pp. 727–734.

[12] G. Zhao, S. Ding, and R. Gutiérrez-Osuna, "Converting foreign accent speech without a reference," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 29, pp. 2367–2381, 2021.

[13] S. Ding, G. Zhao, and R. Gutiérrez-Osuna, "Accentron: Foreign accent conversion to arbitrary non-native speakers using zero-shot learning," *Comput. Speech Lang.*, vol. 72, p. 101302, 2022.

[14] K. Sadov, M. Hutter, and A. Near, "Low-latency real-time voice conversion on CPU," *arXiv preprint arXiv:2311.00873*, Nov. 2023.

[15] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, M. Zeng, X. Yu, and F. Wei, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE J. Sel. Topics Signal Process.*, 2022.

[16] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. NeurIPS*, 2020.

[17] R. E. Kass and A. E. Raftery, "Bayes factors," *J. Amer. Statist. Assoc.*, vol. 90, no. 430, pp. 773–795, 1995.